

AI3602 Data Mining Homework 2.1

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Question 1

In this case, with random teleports, we suppose we follow a randomlink with probability β and **teleport with probability** $1 - \beta$. The transition matrix is

$$A = \beta M + (1 - \beta) \frac{1}{n+1} \quad (1)$$

where

$$M = \begin{bmatrix} 0 & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & 0 \\ \frac{1}{n} & 0 & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & 0 \\ \frac{1}{n} & \frac{1}{n} & 0 & \frac{1}{n} & \cdots & \frac{1}{n} & 0 \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & 0 & \cdots & \frac{1}{n} & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & 0 & 0 \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & 1 \end{bmatrix}, \quad (2)$$

Suppose spider traps absorb all importance.

To calculate the close form of the PageRank vector, we need to solve the equation

$$r = A \cdot r, \quad (3)$$

and that is

$$r_j = \beta \sum_{i=1}^{n+1} M_{ji} r_i + \frac{1 - \beta}{n+1}, \quad (4)$$

$$r = \beta M r + \frac{1 - \beta}{n+1}. \quad (5)$$

According to the symmetry of the graph, $r_1 = r_2 = \cdots = r_n$, and we only need to solve r_1 and r_{n+1} .

$$\begin{cases} r_1 = \beta \sum_{i=2}^n \frac{1}{n} \cdot r_1 + \frac{1 - \beta}{n+1} \\ r_{n+1} = \beta \sum_{i=1}^n \frac{1}{n} \cdot r_1 + \beta r_{n+1} + \frac{1 - \beta}{n+1} \end{cases} \quad (6)$$

$$r_1 = \frac{\frac{1 - \beta}{n+1}}{1 - \beta \frac{n-1}{n}} = \frac{n(1 - \beta)}{n(n+1) - \beta(n+1)(n-1)} = r_i, i = 2, 3, \dots, n \quad (7)$$

$$r_{n+1} = \frac{\beta r_1 + \frac{1 - \beta}{n+1}}{1 - \beta} = \frac{\beta r_1}{1 - \beta} + \frac{1}{n+1} = \frac{n\beta}{n(n+1) - \beta(n+1)(n-1)} \quad (8)$$

If spider traps absorb all importance and we use naive teleport where surfer teleport out with equal probability, the transition matrix is

$$M = \begin{bmatrix} 0 & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & \frac{1}{n+1} \\ \frac{1}{n} & 0 & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & \frac{1}{n+1} \\ \frac{1}{n} & \frac{1}{n} & 0 & \frac{1}{n} & \cdots & \frac{1}{n} & \frac{1}{n+1} \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & 0 & \cdots & \frac{1}{n} & \frac{1}{n+1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & 0 & \frac{1}{n+1} \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & \frac{1}{n+1} \end{bmatrix}. \quad (9)$$

Similarly,

$$\begin{cases} r_1 = \beta \sum_{i=2}^n \frac{1}{n} \cdot r_1 + \frac{\beta}{n+1} r_{n+1} + \frac{1-\beta}{n+1} \\ r_{n+1} = \beta \sum_{i=1}^n \frac{1}{n} \cdot r_1 + \frac{\beta}{n+1} r_{n+1} + \frac{1-\beta}{n+1} \end{cases} \quad (10)$$

$$r_1 = \frac{n}{n^2 + n + \beta} = r_i, i = 2, 3, \dots, n \quad (11)$$

$$r_{n+1} = \frac{n + \beta}{n^2 + n + \beta} \quad (12)$$

Question 2

In this case, suppose the teleport probability to A is 0.2 instead of 0.8. Similarly

$$A = \beta M + (1 - \beta)T \quad (13)$$

where

$$M = \begin{bmatrix} 0 & \frac{1}{2} & 1 & 0 \\ \frac{1}{3} & 0 & 0 & \frac{1}{2} \\ \frac{1}{3} & 0 & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & 0 & 0 \end{bmatrix}, \quad (14)$$

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad (15)$$

The equations are:

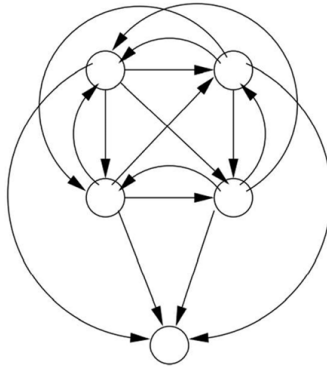
$$\begin{cases} r_1 = \frac{2}{5}r_2 + \frac{4}{5}r_3 + 0.2(r_1 + r_2 + r_3 + r_4) \\ r_2 = \frac{4}{15}r_1 + \frac{2}{5}r_4 \\ r_3 = \frac{4}{15}r_1 + \frac{2}{5}r_4 \\ r_4 = \frac{4}{15}r_1 + \frac{2}{5}r_2 \end{cases}, \quad (16)$$

$$\begin{cases} r_1 = \frac{3}{7} \\ r_2 = \frac{4}{21} \\ r_3 = \frac{4}{21} \\ r_4 = \frac{4}{21} \end{cases} \quad (17)$$

Original Questions

Q1. Cliques and Dead Ends

Consider a Web consisting of a *clique* (i.e., a graph such that every two different nodes are adjacent) of n nodes and a single additional node that is a successor of each of the n nodes in the clique. Below shows the Web for $n = 4$.



Determine the PageRank for each page, as a function of n and β , where n is the number of nodes, and β is the probability of random teleport.

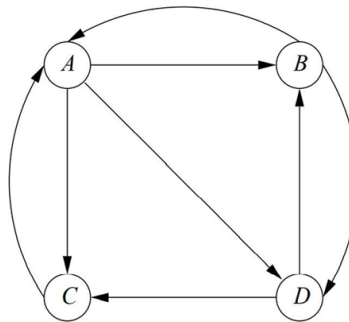
Figure 1: Question 1

Reference

None.

Q2. Topic-Specific PageRank

Compute the topic-specific PageRank for the graph below. Assume the teleport set is $S = \{A\}$ and the teleport probability $\beta = 0.8$.



Note. While a mathematical derivation is preferred, you are also allowed to use `numpy` or similar numerical linear algebra packages to solve the eigenvalue decomposition and provide numerical results. This will not affect your final score.

Figure 2: Question 1